STACKS, DISCRIMINANTS: LEMMA-NOETHER-PAIRING-FLAT-QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1427-1432 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1434-1482, the complete author proof (47 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_discriminant.json (pb=2114, stmt 1427-1432, proof 1434-1482).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00HN: sources/stacks-discriminant/pinned-original-source.tex, source lines 1482–1482. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00UJ: sources/stacks-discriminant/pinned-original-source.tex, source lines 1482–1482. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT3: sources/stacks-discriminant/pinned-original-source.tex, source lines 450–454. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT5: sources/stacks-discriminant/pinned-original-source.tex, source lines 493–497. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVM: sources/stacks-discriminant/pinned-original-source.tex, source lines 1311–1320. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVL: sources/stacks-discriminant/pinned-original-source.tex, source lines 1295–1303. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVR: sources/stacks-discriminant/pinned-original-source.tex, source lines 1386–1411. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVP: sources/stacks-discriminant/pinned-original-source.tex, source lines 1362–1369. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a flat quasi-finite homomorphism of Noetherian rings. The pairing of Remark 0BVP induces an isomorphism $J \to \mathrm{Hom}_B(\omega_{B/A}, B)$.*

Proof. We first prove this when $A \to B$ is finite and flat. In this case we can localize on A and assume B is finite free as an A-module. Let $b_1, \ldots, b_n$ be a basis of B as an A-module and denote $b_1^\vee, \ldots, b_n^\vee$ the dual basis of $\omega_{B/A}$. Note that $\sum b_i \otimes c_i \in J$ maps to the element of $\mathrm{Hom}_B(\omega_{B/A}, B)$ which sends $b_i^\vee$ to c_i. Suppose $\varphi : \omega_{B/A} \to B$ is B-linear. Then we claim that $\xi = \sum b_i \otimes \varphi(b_i^\vee)$ is an element of J. Namely, the B-linearity of φ exactly implies that $(b \otimes 1)\xi = (1 \otimes b)\xi$ for all $b \in B$. Thus our map has an inverse and it is an isomorphism.

Let $\mathfrak{q} \subset B$ be a prime lying over $\mathfrak{p} \subset A$. We will show that the localization

$$J_{\mathfrak{q}} \longrightarrow \mathrm{Hom}_B(\omega_B/A, B)_{\mathfrak{q}}$$

is an isomorphism. This suffices by Algebra, Lemma 00HN. By Algebra, Lemma 00UJ we can find an étale ring map $A \to A'$ and a prime ideal $\mathfrak{p}' \subset A'$ lying over $\mathfrak{p}$ such that $\kappa(\mathfrak{p}') = \kappa(\mathfrak{p})$ and such that

$$B' = B \otimes_A A' = C \times D$$

with $A' \to C$ finite and such that the unique prime $\mathfrak{q}'$ of $B \otimes_A A'$ lying over $\mathfrak{q}$ and $\mathfrak{p}'$ corresponds to a prime of C. Let J' be the annihilator of $\mathrm{Ker}(B' \otimes_{A'} B' \to B')$. By Lemmas 0BT3, 0BVM, and 0BVR the map $J' \to \mathrm{Hom}_{B'}(\omega_{B'/A'}, B')$ is gotten by applying the functor $- \otimes_B B'$ to the map $J \to \mathrm{Hom}_B(\omega_{B/A}, B)$. Since $B_{\mathfrak{q}} \to B'_{\mathfrak{q}'}$ is faithfully flat it suffices to prove the result for $(A' \to B', \mathfrak{q}')$. By Lemmas 0BT5, 0BVL, and 0BVR this reduces us to the case proved in the first paragraph of the proof. □